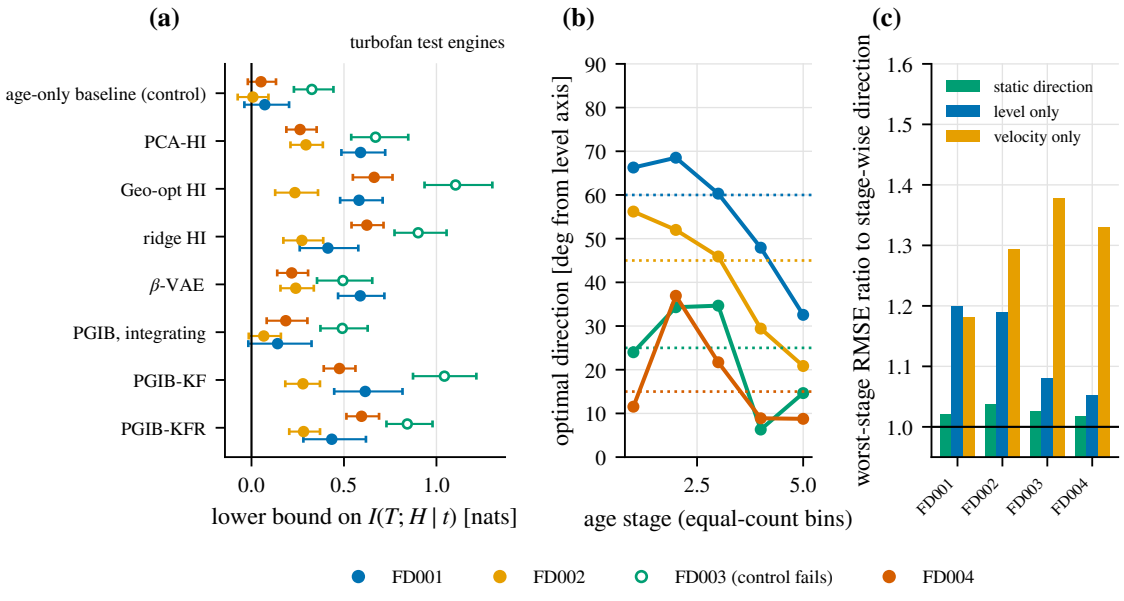
(a)
turbofan test engines
age-only baseline (control)
PCA-HI
Geo-opt HI
ridge HI
β-VAE
PGIB, integrating
PGIB-KF
PGIB-KFR
0.0
0.5
1.0
lower bound on I(T; H | t) [nats]
(b)
optimal direction [deg from level axis]
90
80
70
60
50
40
30
20
10
0
2.5
5.0
age stage (equal-count bins)
(c)
worst-stage RMSE ratio to stage-wise direction
1.6
1.5
1.4
1.3
1.2
1.1
1.0
static direction
level only
velocity only
FD001
FD002
FD003
FD004
FD001
FD002
FD003 (control fails)
FD004